

Forces create a clockwise moments.

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B

Loss in KE = Gain in EPE

$$\left(\frac{1}{2}mv^2 + \frac{1}{2}mv^2\right) - 0 = \frac{Qq}{4\pi\epsilon_0 r} - 0$$

$$r = \frac{Qq}{4m\pi\epsilon_0 v^2}$$

$$\begin{aligned} r &= \frac{(1.60 \times 10^{-19})^2}{4(9.11 \times 10^{-31})\pi(8.85 \times 10^{-12})(2.0 \times 10^6)^2} \\ &= 6.3 \times 10^{-10} \text{ m (2sf)} \end{aligned}$$